IRFAN KHALIL

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SUMMARY

Process/Operations Engineer (petrochemical, hydrogen peroxide) with more than 14 years' experience in petrochemical and hydrogen peroxide(H_2O_2) plants, specializing in Anthraquinone Auto-Oxidation (AO) processing, commissioning, pre-commissioning, start/stop of the plants, and optimizing of the processes. Qualified in Process Safety Management, HAZOP analysis, PSSR, Permit to Work, and able to deliver high-quality projects in petrochemical and hydrogen peroxide processing within the scope of PSM, HAZOP analysis, DCS operation, problem-solving.

Proven success in managing process safety, operations, and projects in petrochemical, hydrogen peroxide,processing. Experienced in operating plants under extreme temperatures and pressure, working in high-risk environments, and handling highly hazardous substances. Extensive experience in emergency response and incident management, with excellent safety performance (zero HSE incidents).

KEY SKILLS

- Process Safety Management (PSM), HAZOP, PSSR
- Permit to Work (PTW), LOTO, JSA
- Refinery & Petrochemical Operations
- Commissioning & Pre-Commissioning
- Startup & Shutdown Operations
- Emergency Response & Incident Handling
- DCS (Distributed Control System) Operation
- Root Cause Analysis (RCA)
- Troubleshooting & Process Optimization
- Preventive Maintenance
- Risk Assessment
- KPI Monitoring

EDUCATION

PBTE – Multan Swedish Institute of Technology
DAE (Chemical Technology)

Built a strong technical foundation in chemical engineering principles, process plant operations, and industrial safety standards. Acquired hands-on understanding of equipment handling, process control, and troubleshooting, forming the basis for a successful career in petrochemical and hydrogen peroxide plant operations

BISE D.G Khan
Matriculation (Science)

EXPERIENCE

Assistant Operations Engineer (Hydrogen Peroxide Plant – HPO)

Dec 2022
Present

Engro Polymer & Chemicals Ltd

- Working at Hydrogen Peroxide (HPO) Plant

• Executed erection, pre-commissioning, and commissioning activities of Hydrogen Peroxide (HPO) Plant in accordance with P&IDs, drawings, and technical specifications

• Prepared and implemented pre-commissioning and commissioning checklists ensuring system readiness and compliance

• Conducted equipment inspections and functional testing of pumps, valves, transmitters, analyzers, and safety interlocks

• Performed flushing, cleaning, and line clearing activities ensuring system integrity and contamination-free operations

• Supported JSA and HAZOP studies to develop and implement safe work practices and operational procedures

• Ensured compliance with Process Safety Management (PSM), Permit to Work (PTW), and HSE standards

• Performed plant startup and shutdown operations maintaining process stability and safe operations

• Executed system checkout and commissioning activities ensuring smooth plant startup

• Monitored and controlled plant operations using DCS and field instrumentation within defined operating limits

• Operated and maintained key process equipment including Hydrogenator, Oxidizer, Extractor, WS Regeneration System, SRU, Distillation, and Concentration Units

• Handled catalyst loading/unloading and candle filter backwashing activities

• Managed Hydrogen Peroxide production (35%–50% H_2O_2) ensuring product quality and process efficiency

• Identified and resolved process upsets through troubleshooting, minimizing downtime and improving reliability

• Trained operators and supported team development to enhance operational efficiency

• Participated in Emergency Response Team (ERT) activities ensuring quick and safe emergency handling

• Maintained accurate log sheets, operational records, and documentation

• Coordinated with maintenance, instrumentation, and engineering teams for smooth plant operations

Field Operator **Jan 2015 – Nov 2022**

Descon Oxychem Ltd *Sheikhupura*

- Hands-on experience in Hydrogen Peroxide (H_2O_2) production using Anthraquinone Auto-Oxidation (AO) Process, including hydrogenation, oxidation, extraction, and working solution regeneration systems
- Controlled critical process parameters and responded to process upsets and abnormalities
- Optimized plant performance through process monitoring and troubleshooting
- Ensured strict compliance with SOPs, safety procedures, and operational standards
- Handled key equipment including Reactors, Extractor Columns, SRU, and Distillation

Production Officer / Field Operator **Nov 2014 – Dec 2015**

Nimir Industrial Chemicals Ltd *Sheikhupura*

Worked as a Production Officer / Field Operator in the Fatty Acid Plant, responsible for the safe and efficient operation of key process equipment, including Splitting Columns and Distillation Columns. Operated and monitored the fatty acid splitting process, where natural oils and fats were hydrolyzed to produce crude fatty acids and crude glycerine. Controlled critical process parameters to ensure efficient separation and purification of fatty acids and glycerine, maintained stable plant operation, optimized product quality and process efficiency, and

ensured compliance with standard operating procedures (SOPs)

Field Operator

Rupafil Ltd **Sep 2012 – Oct 2014**

Sheikhupura

Worked as a Field Operator in the Utility plant responsible for the safe and efficient operation of critical utility systems supporting continuous plant operations. Operated and monitored the Demineralized (DM) Water Treatment Plant to ensure a consistent supply of high-quality demineralized water for process use. Managed Cooling Water Systems and Cooling Towers to maintain optimum cooling water circulation and heat removal for process equipment. Operated and monitored Air Compressors to ensure a reliable supply of compressed air for plant utilities and instrumentation. Performed routine field inspections, monitored operating parameters, responded to equipment abnormalities, coordinated with the control room during operational activities, and ensured compliance with SOPs

CERTIFICATIONS

- Process Safety Management (PSM)
  Engro polymer and chemical limited (EPCl)
- Certified Permit to Work System
  Engro polymer and chemical limited (EPCl)
- Emergency response training *(ERT)*
  Engro polymer and chemical limited (EPCl)
- Fire Safety & Emergency Drills
  Engro polymer and chemical limited (EPCl)
- CPR & AED Training
  American heart association (AED)
- Basics of fire safety
  Cambridge international qualifications uniathena.
- Introduction to MS Excell
  simply learn skill up
- Basic problem solving
  KSBL leap training
- *Team work and collaboration*
  KSBL leap training

KEY ACHIEVEMENTS

- Executed multiple plant startups and shutdowns without safety incidents
- Maintained zero major HSE incidents over 14-year career”
- Contributed to safe commissioning and successful startup of HPO plant
- Active participation in emergency response drills and fire safety operations
- Demonstrated strong capability in incident handling and operational risk control
- Honored with an Achievement Award for excellence in executing two Cold Start-ups and one Hot Start-up while ensuring safe and reliable plant operations.
- Received an Achievement Award for promptly identifying a nitric acid leakage in the chemical warehouse and ensuring its safe handling, preventing a potential safety incident.
- Received 2 consecutive HSE Excellence Awards at EPCL”
- Identified nitric acid leakage and prevented potential plant shutdown

TECHNICAL EXPERTISE

- **Process;** Hands-on Operation of Hydrogen Peroxide (H_2O_2) Production Plant (Anthraquinone Auto-Oxidation Process)
- **System;** Practical Experience with DCS (Distributed Control System) & Field Instrumentation
- **Equipment;** Hands-on Operation and Monitoring of Pumps, Compressors, Heat Exchangers, Distillation Columns, Reactors, and Extraction Systems
- Process Monitoring, Troubleshooting, Process Optimization & Performance Improvement
- Maintenance Coordination, Plant Shutdown/Startup Execution & Equipment Isolation.
- Safe Operation of Process Equipment in Compliance with Process Safety Management (PSM).
- Field Inspection, Process Parameter Monitoring & Operational Decision Making

Key Responsibilities

- Monitor and control critical process parameters to ensure safe, stable, and efficient plant operations.
- Execute plant startup, normal shutdown, and emergency shutdown procedures in accordance with established operating standards.
- Supervise and coordinate maintenance activities, including Hot Work, Confined Space Entry, Line Breaking, and equipment isolation, while ensuring full compliance with safety requirements.
- Maintain accurate operational log sheets, shift reports, equipment checklists, and process records.
- Coordinate effectively with Maintenance, Instrumentation, Electrical, and Process Engineering teams to ensure smooth plant operation and timely issue resolution.
- Ensure strict compliance with HSE policies, Process Safety Management (PSM) requirements, Permit-to-Work (PTW) procedures, and company safety standards.

Personal Details

Date of Birth: 19 Sep 1990

Nationality: Pakistani

Marital Status: Married

Address: Muzaffargarh, Pakistan

Languages

English (professional working proficiency B-2 level), Urdu (fluent), Saraiki (Native)